



Beyond algorithm aversion: The impact of psychological readiness on algorithmic advice

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ARTICLE INFO

Keywords:

Mindset
Algorithm aversion
Algorithm appreciation
Uncertainty
Psychological readiness
Advice-taking
Trust

ABSTRACT

While algorithms increasingly outperform human experts and gain widespread adoption, many individuals still resist using them due to algorithmic aversion. Although prior research has examined the appreciation and avoidance of algorithmic advice, the underlying mechanisms driving these decisions remain underexplored. This paper investigates the role of individuals' readiness to act, specifically whether they adopt a deliberative or implemental mindset, in shaping their openness to algorithmic advice. Across three hypothetical studies and one incentive-compatible study, results show that individuals in a deliberative mindset, characterized by thoughtful evaluation, tend to prefer advice from human sources. In contrast, those in an implemental mindset, characterized by action-oriented thinking, are more likely to prefer algorithmic advice. Additionally, the findings reveal that perceived uncertainty moderates the influence of mindset on algorithmic receptiveness. These findings offer nuanced insights into the psychological mechanisms that drive engagement with algorithms and suggest practical strategies to enhance collaboration with both algorithmic and human recommendations.

Responses to algorithms often exhibit ambivalence. Many individuals acknowledge their efficiency and objectivity, yet a significant number remain skeptical and favor human judgment. For instance, people consult financial advisors for investment strategies and guidance, while others trust algorithmic tools to carry out their decisions. This divergence illustrates a fundamental dilemma in decision-making: when do individuals lean toward algorithmic versus human recommendations? Understanding the conditions that drive this decision is crucial for designing more effective advisory systems.

Algorithms are rule-based computational procedures that transform data inputs into decision outputs, enabling them to detect meaningful patterns and produce results with speed and precision (Gillespie, 2014; Mahmud et al., 2022). As these systems continue to expand, their involvement in decisions traditionally guided by human judgment is increasing (De Bruyn et al., 2020; Mahmud et al., 2022). This shift has led to a growing phenomenon known as algorithm appreciation, where individuals tend to prefer algorithmic input over human advice. For example, Logg et al. (2019) found that people were more influenced by recommendations they believed originated from algorithms, even when the quality of advice was equivalent. Similar patterns have emerged across various domains where algorithms are viewed as more objective and trustworthy, including financial planning (Cao et al., 2024), education (Kim & Koo, 2024), and healthcare (Chusteck, 2024).

Despite these positive perceptions, many people remain wary of algorithms. Even when algorithms outperform human experts, they are not always readily accepted (Castelo et al., 2019; Dietvorst et al., 2015). This phenomenon, known as algorithm aversion, reflects a consistent tendency for people to favor human judgment over algorithmic advice, even when the latter proves more accurate (Dietvorst et al., 2015). This skepticism is often driven by concerns about loss of control, lack of accountability, and the belief that algorithms cannot adequately handle context specific nuances (Cadario et al., 2021; Longoni et al., 2019; Sharan & Romano, 2020). For instance, encountering a single algorithmic error can cause people to revert to human decision makers, despite the algorithm's overall stronger performance (Dietvorst et al., 2015).

While research on algorithm appreciation and aversion has expanded, understanding of why and when these responses occur remains limited. Much of the existing work focuses on external factors such as design features or situational triggers; for example, people are more likely to reject algorithms after experiencing inaccuracies or perceiving systems as impersonal or inflexible (Castelo et al., 2019; Dietvorst et al., 2015). Other studies highlight dispositional characteristics, including personality traits and risk perceptions (Araujo et al., 2020; Sharan & Romano, 2020). More recently, frameworks such as the Capability and Personalization Model suggest that algorithm preference

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<https://doi.org/10.1016/j.chb.2025.108824>

Received 16 May 2025; Received in revised form 4 October 2025; Accepted 6 October 2025

Available online 10 October 2025

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increases when algorithms are seen as more capable than humans and when personalization is considered unnecessary (Qin et al., 2025). Although these contributions provide important insights, they pay little attention to a critical dimension: the role of individuals' psychological states during decision making.

This gap is especially important because a person's motivational orientation at the moment of choice can significantly shape their openness to algorithmic advice. Drawing on mindset theory (Gollwitzer, 1990, 1999), the present research addresses this gap by examining how psychological readiness to act affects responses to algorithmic recommendations. This work finds that individuals in a deliberative mindset, who are still evaluating alternatives, tend to be more open to diagnostic input and actively seek information from human sources. In contrast, those in an implemental mindset, having committed to a course of action, are more likely to be receptive to algorithmic advice. This research also shows that the effect is particularly pronounced in high-stakes contexts. Consequently, the same algorithmic recommendation may be interpreted and acted upon differently depending not only on its content or origin but also on the decision maker's underlying motivational state.

Theoretically, this work advances understanding by moving beyond explanations that focus on static individual traits or surface-level system features. Instead, it emphasizes the critical role of dynamic motivational factors, specifically people's psychological readiness during decision making, in shaping their receptivity to algorithmic input. This approach bridges gaps between cognitive and motivational theories by demonstrating that mindset states influence both information processing and the trust and acceptance of algorithmic recommendations. By capturing this interplay, the research challenges existing models that treat algorithm aversion or appreciation as fixed preferences and instead suggests that openness to algorithms varies systematically with internal goal states and decision phases. Moreover, the framework offers conceptual clarity by distinguishing between deliberative and implemental mindsets as key drivers in human algorithm interaction, enriching theoretical models that have largely overlooked temporal and motivational dynamics.

From a practical standpoint, understanding these motivational mechanisms offers valuable guidance for the timing and delivery of algorithmic recommendations. Systems that tailor their suggestions based on users' decision phase may enhance trust and adoption. More broadly, this research underscores the need for a psychologically informed approach to algorithm design, one that respects the fluid and context-dependent nature of human thought and motivation. This perspective refines theories of human-algorithm interaction and advances the development of more adaptive, human-centered algorithmic technologies.

1. Theoretical background

The integration of algorithms into decision making represents a transformative shift in how problems are solved across institutions and everyday life. Rather than relying solely on human intuition or manual processing, modern systems increasingly depend on algorithmic instructions, which are defined as structured computational procedures that transform raw data into actionable outputs (Gillespie, 2014). Innovations in artificial intelligence, including techniques like machine learning and natural language processing, have driven the widespread adoption of these technologies, allowing algorithms to operate across numerous industries with growing autonomy. Leading firms such as Accenture have demonstrated this shift by investing billions into AI-powered infrastructure designed to streamline operations and support data-driven strategic planning (Accenture, 2023).

As these systems become more advanced, they assist human decision makers in routine tasks and, in many cases, replace them in complex ones. Algorithms built on statistical models can analyze patterns in large datasets and produce decisions independently, without explicit human guidance (Lee, 2018). Whether applied to objective judgments or

context-rich evaluations, algorithms are increasingly shaping decisions that impact individuals, businesses, and society at large (Mahmud et al., 2022).

1.1. Evolving reliance in algorithmic advice

Over the past several decades, researchers have shown a growing tendency for individuals to prefer algorithmic recommendations over human judgment across a range of domains (Cao et al., 2024; Kim & Koo, 2024; Logg et al., 2019). This phenomenon, known as algorithm appreciation, reflects increasing trust in machine-based input (Logg et al., 2019). Its origin traces back to Meehl's (1954) foundational study, which demonstrated that simple statistical models could outperform clinicians in making medical diagnoses. Since then, a large body of research has confirmed the preference for algorithms in numerous settings, including business (Huang & Rust, 2021), healthcare (Chustecki, 2024; Ngiam & Khor, 2019), and human resource management (Meijerink et al., 2021).

This preference extends beyond algorithms' superior performance. Individuals often perceive these systems as more impartial and trustworthy. Algorithms leveraging big data reduce human bias by limiting subjective judgments typically involved in evaluating others (Newman et al., 2020). Supporting this view, You, Yang, and Li (2022) found that people tend to trust advice labeled as algorithm-generated more than identical advice attributed to humans. This trust increases further in contexts where people must choose between human and AI recommendations. Moreover, as task complexity grows, individuals increasingly rely on algorithms rather than social cues (Bogert et al., 2021).

Trust in algorithmic recommendations is influenced not only by task characteristics but also by individual self-perceptions, such as self-evaluation (Araujo et al., 2020; Mahmud et al., 2022), and by personality traits like locus of control (Sharan & Romano, 2020). Additionally, in situations characterized by complexity or information overload, people often turn to algorithmic guidance, valuing its ability to efficiently process large volumes of data (Goddard et al., 2012; Siegel et al., 2024; Stenzl et al., 2022). These findings suggest that algorithm appreciation reflects perceived competence as well as deeper considerations related to control.

Despite this growing trust, ambivalence remains. While many acknowledge the strengths and advantages of algorithms, they hesitate to fully entrust them as decision-makers. This tension between acceptance and skepticism has led to a substantial body of research on algorithm aversion, revealing the complex and sometimes contradictory attitudes people hold toward these technologies.

1.2. The psychology of resisting algorithms

While algorithms are often trusted and valued, they frequently encounter resistance. This phenomenon, known as algorithm aversion, describes a preference for human judgment over algorithmic input, even in the face of evidence that algorithms perform just as well or better (Berger et al., 2021; Dietvorst et al., 2015). Many individuals continue to rely on the intuition of human experts or their own judgment rather than follow the recommendations produced by automated systems (Burton et al., 2020). For some, this resistance is driven by a perceived distrust of algorithms, irrespective of their objective accuracy (Mahmud et al., 2022; Prah & Van Swol, 2017). Such skepticism spans multiple domains. For example, in healthcare, patients often report greater comfort with human providers than with algorithmic diagnostics (Longoni et al., 2019), while recruitment professionals tend to favor their own expertise over algorithmic recommendations (Highhouse, 2008; Lacroux & Martin-Lacroux, 2022). These findings suggest that performance alone does not determine acceptance; rather, a combination of psychological and perceptual factors influences how people respond to algorithmic input.

A key psychological barrier involves the perception that algorithms

are inherently rigid, unable to adapt or incorporate new information in context-specific ways (Castelo et al., 2019; Highhouse, 2008; Mahmud et al., 2022). Algorithms are often seen as impersonal or detached, which can be problematic in settings that call for empathy or human understanding (Grove & Meehl, 1996; Longoni et al., 2019). Feeling reduced to a mere data point or statistical abstraction can elicit discomfort and mistrust, especially in emotionally sensitive or high-stakes contexts (Önköl et al., 2009). This discomfort is often amplified by the opacity of many algorithmic processes, which contrasts with the perceived transparency of human reasoning (Goodwin et al., 2013; Mahmud et al., 2022; van Dongen & van Maanen, 2013). When algorithmic outcomes deviate from expectations and lack clear explanations, skepticism and rejection frequently follow (Goodwin et al., 2013).

Recent research has also investigated dispositional factors underlying these responses. A growing body of work highlights general algorithm aversion, which involves habitual rejection of algorithmic input even without critical evaluation (Mahmud et al., 2022). This aversion often arises from heuristics or perceptions that algorithms lack empathy, adaptability, or nuanced judgment (Luo et al., 2019). Motivational factors also influence algorithm aversion. When decisions are weakly tied to personal or financial consequences, individuals tend to be less willing to rely on algorithmic recommendations (van Esch, 2024). Additionally, people may resist algorithmic input when it threatens their sense of autonomy or intrinsic motivation, as they prefer to maintain control over decisions that reflect their personal values and goals (Sankaran et al., 2021).

Despite growing scholarly interest in how individuals respond to algorithmic systems, existing findings on algorithm appreciation and aversion remain conceptually fragmented. On one hand, some studies emphasize the benefits of algorithmic input, highlighting its accuracy, objectivity, and consistency (Cao et al., 2024; Legg et al., 2019). On the other hand, people often remain reluctant to follow algorithmic recommendations, despite evidence that these recommendations can surpass human judgment (Dietvorst et al., 2015; Longoni et al., 2019). This raises an important question: when and why do people accept or reject algorithmic recommendations?

To address this puzzle, prior research has explored a range of psychological factors. Kopecka et al. (2024) found that individuals with an analytical thinking style preferred structured, rule-based algorithmic explanations, whereas those with a holistic thinking style favored intuitive, narrative-driven outputs. This suggests that acceptance depends, in part, on the perceived alignment between an algorithm's explanatory format and the user's cognitive orientation. Similarly, Longoni and Cian (2022) showed that goal type influences receptivity: individuals were more open to algorithmic input when pursuing practical or utilitarian goals but preferred human advice for hedonic or emotionally motivated decisions. Building on the idea that goals guide technology use, Mahmud et al. (2023) examined technology readiness, defined as a person's inclination to adopt new technologies for their personal or professional goals. They showed that higher technology readiness, especially when linked to aims, predicts greater willingness to rely on algorithmic input.

Although prior research has examined the connection between goal-directed motivation and technology acceptance, it remains unclear how this relationship plays out in the moment of decision-making. When individuals encounter algorithmic recommendations, their current motivational orientation may differ. Some may be contemplating various options, while others may already be committed to a specific course of action. These differences in psychological readiness are likely to shape how individuals perceive and respond to algorithmic input. Building on this idea, the present research proposes how receptivity to algorithmic versus human recommendations depends on whether individuals are in a deliberative or implemental mindset at the time of the decision.

1.3. Deliberative versus implemental mindsets

Gollwitzer's (1990) mindset theory proposes two distinct phases of goal pursuit, each linked to a different mindset. In the pre-decisional phase, individuals assess various goals by weighing their pros and cons. This phase is characterized by a deliberative mindset, where individuals focus on carefully evaluating options and are not yet prepared to take action. They concentrate on the potential benefits and drawbacks of each goal before making a decision. Once a decision is made, individuals shift to an implemental mindset to facilitate goal execution. At this stage, they are prepared to act and concentrate on the practical aspects of implementing the chosen goal, with less emphasis on ongoing evaluation (Gollwitzer, 1990, 1999; Kim, 2021). Deliberative mindsets enhance attention to and recall of outcome-related information, whereas implemental mindsets improve focus on and recall of execution-related details. Importantly, these mindsets can extend beyond the initial task, carrying over to influence subsequent tasks and demands (Gollwitzer, 1999; Hamilton et al., 2011).

These distinct mindsets play a crucial role in shaping how individuals approach external information and guidance during decision-making processes. Because they influence the focus and cognitive resources devoted to evaluating options, mindsets affect not only what information people attend to but also how they interpret and value input from different sources (Kim & Häubl, 2021; Murphy & Dweck, 2016). The mental framework established by either mindset creates a context in which individuals filter incoming recommendations, potentially altering their openness and responsiveness to advice depending on its origin. In this way, mindset may operate as a lens through which algorithmic and human input is perceived and integrated into the decision-making process.

2. Hypotheses development

Building on prior research and the theoretical framework of deliberative versus implemental mindsets, the study hypothesizes that individuals in a deliberative mindset will respond more favorably to human-generated recommendations than to those generated by algorithms. In this mindset, people tend to engage in systematic processing, carefully evaluating options and seeking guidance that is comprehensive and tailored to their needs. Human recommendations may be especially persuasive in this context, as they often provide qualitative insights and experiential knowledge that align with the desire for deeper understanding and personalized feedback.

However, when individuals are in an implemental mindset, characterized by a focus on action and goal completion, algorithm-based recommendations may carry greater influence. People in this mindset tend to prioritize efficiency and practicality, seeking clear, actionable solutions that facilitate goal attainment (Gollwitzer, 1990). Research by Henderson, De Liver, and Gollwitzer (2008) suggests that implemental mindsets promote more accessible and well-defined attitudes toward goals, which can enhance preferences for objective information. Algorithmic recommendations are often perceived as more objective than human recommendations (Henestrosa et al., 2023; Kim & Koo, 2024), and their structured, data-centric nature aligns well with the action-oriented and efficiency-driven orientation of implemental thinkers. As a result, algorithm-based recommendations may be particularly persuasive in decision-making contexts shaped by an implemental mindset.

H1A. Individuals in a deliberative mindset respond more favorably to human-generated recommendations than to algorithm-generated ones.

H1B. Individuals in an implemental mindset respond more favorably to algorithm-generated recommendations than to human-generated ones.

Among the various factors contributing to algorithm aversion, this

study proposes uncertainty as a significant moderator. Previous studies have indicated that in volatile and unpredictable decision environments, individuals are more inclined to reject even highly accurate algorithms (Dietvorst & Bharti, 2020). For example, in forecasting, people often trust human predictions more than algorithmic ones, especially when they believe the advice comes from an expert rather than a statistical method (Kawaguchi, 2021; Önköl et al., 2009). Perkins et al. (2010) showed that in high-risk contexts, individuals tend to prefer human advisors over algorithmic solutions. This reluctance is driven by concerns about potential consequences and uncertainty regarding outcomes, which leads people to be less willing to delegate critical decisions to algorithms. However, this finding is not always conclusive. Sutherland et al. (2016) argue that in uncertain advising environments, people are more inclined to rely on algorithms because of their perceived expertise and accuracy.

Given the mixed findings in prior research on individuals' acceptance or rejection of algorithmic advice and the underlying mechanisms in uncertain contexts, this study aims to explore how appreciation or aversion to algorithms manifests in such situations. Specifically, this research proposes that in high-uncertainty situations, individuals in a deliberative mindset are likely to prefer human recommendations. This mindset prioritizes the thorough consideration of potential outcomes, prompting individuals to seek guidance that offers a more comprehensive and context-specific understanding. In such situations, human recommendations are perceived as more reliable because they provide not only human insight but also the nuanced judgment that reflects the complexity and ambiguity inherent in high-risk decisions. In contrast, individuals in an implemental mindset are likely to prefer algorithmic recommendations. The clear, structured nature of algorithmic advice, driven by data and objective criteria, enables more efficient decision-making, particularly in high-risk contexts. Individuals in this mindset would value the simplicity and clarity that algorithms offer, as they allow for quicker, more confident actions when the uncertainty of a situation may otherwise create hesitation.

On the other hand, in low-uncertainty contexts, this study proposes that the relationship between mindset and recommendation preference will shift. Individuals in an implemental mindset are more likely to favor human recommendations over algorithmic ones. With lower risks and stakes, the perceived difficulty of the decision decreases, allowing individuals to prioritize advice that aligns with their values and goals instead of focusing solely on proficiency—something human advisors are better equipped to provide. As individuals in an implemental mindset focus on successfully executing their decisions, they are more likely to value human guidance that combines practical knowledge with emotional insight, making it more adaptable and relevant in situations where decision-making demands a nuanced and context-sensitive approach.

In contrast, individuals in a deliberative mindset prioritize careful analysis and thorough evaluation, even in low-uncertainty contexts. With a mindset oriented toward exploring possibilities and understanding complexities to achieve their goals, these individuals are more open to and appreciative of analytically rigorous suggestions (Fujita et al., 2007). They are likely to favor algorithmic suggestions because such recommendations emphasize rationality and align well with their preference for analytical, evidence-based decision-making.

H2A. In high-uncertainty contexts, individuals in an implemental (vs. deliberative) mindset exhibit more favorable responses to algorithm-generated (vs. human-generated) recommendations.

H2B. In low-uncertainty contexts, individuals in an implemental (vs. deliberative) mindset exhibit more favorable responses to human-generated (vs. algorithm-generated) recommendations.

3. Study overview

Three hypothetical studies and one incentive-compatible study were

conducted to test the proposed hypotheses. Studies 1A and 1B focused on H1A and H1B, examining how individuals respond to algorithmic versus human recommendations when in deliberative versus implemental mindsets. Study 2 tested H2A and H2B by exploring how these mindsets affect reliance on algorithmic versus human recommendations across contexts of high and low uncertainty. Finally, Study 3 tested H1A and H1B within an incentive-compatible framework, investigating participants' behavioral intentions in real-world scenarios.

This work reports the targeted sample sizes, as well as all conditions and measures. Data were analyzed using SPSS (Version 29). The materials, data, and code for each study are available at https://osf.io/396dk/?view_only=2f92d15cfd4d4be585e073f403956e6b. This research was approved by the Institutional Review Board (IRB) at the author's institution.

4. Study 1A

Study 1A tested H1A and H1B using a between-subjects design, proposing that individuals in a deliberative mindset would respond more favorably to human-generated recommendations. In contrast, individuals in an implemental mindset would respond more favorably to algorithm-generated recommendations.

4.1. Method

Study 1A targeted 260 participants and received responses from 262 U.S.-based participants through Prolific.¹ Participants who failed to answer the attention check question correctly (e.g., "The city councilmen refused the demonstrators a permit because they feared violence. Who feared violence, the councilmen or the demonstrators?") were excluded from the analysis, leaving a final sample of 248 U.S.-based participants ($M_{\text{age}} = 40.96$; 47.6% female, 50.4% male, and 2.0% other). A post hoc power analysis conducted with G*Power revealed that the study achieved 88% power to detect the observed effect size.² Randomization checks revealed no significant differences in age ($p = .250$) or gender ($p = .589$) across conditions, indicating successful random assignment. A 2 (mindset: deliberative versus implemental) $\times$ 2 (recommendation type: algorithm versus human) between-subjects design was used, with participants randomly assigned to one of the four conditions. Preregistration details can be found at: <https://aspredicted.org/txyw-ksw3.pdf>.³

The mindset manipulation aimed to induce either a deliberative or implemental cognitive orientation, based on Gollwitzer's mindset theory (Gollwitzer, 1990), which distinguishes between pre-decisional and post-decisional phases of goal pursuit. Participants first read a scenario in which they considered switching banks. To activate a deliberative mindset, they listed three potential positive and three potential negative consequences of switching banks. For the implemental mindset, participants listed three steps for making the switch, including when, where, and how to do so (see Supplemental Material A).

¹ Prior research indicates that participants recruited through Prolific and CloudResearch provide higher-quality data compared to those from other online survey platforms, such as MTurk (Douglas et al., 2023). Specifically, participants from Prolific and CloudResearch are more likely to pass attention checks, give thoughtful responses, and follow instructions carefully. Accordingly, Studies 1–3 exclusively recruited participants from these two platforms.

² Since an a priori power analysis was not conducted for Studies 1A–3, a post hoc power analysis was performed for each study using G*Power.

³ There is a typographical error in the preregistration, which states that participants completed the manipulation task by listing five potentially positive and five potentially negative consequences of switching banks in the deliberative mindset condition, and five steps necessary to execute the plan in the implemental mindset condition. However, in the current study, participants reported three positive and three negative consequences, as well as three steps, in both conditions.

After the mindset manipulation, participants completed manipulation check items adapted from Gollwitzer and Kinney (1989) to assess mindset activation. These included: “How determined do you feel at the moment with respect to the bank switching decision at hand?” (1 = not determined; 7 = very determined); “Do you feel that you have committed yourself to a certain course of action?” (1 = not committed; 7 = very committed); and “Do you feel that you have committed yourself to utilizing certain occasions or opportunities to act?” (1 = not committed; 7 = very committed).

After completing this task, participants were asked to imagine they were considering a new workout routine but, before committing, they wanted to ensure it was the right decision for them. This scenario was selected because it reflects a familiar, real-life context where individuals often seek advice before making a health-related decision, making it a suitable domain for testing responses to recommendation sources. In the algorithmic recommendation condition, participants were informed that they had discovered a newly launched algorithm-based AI fitness app designed to assess their fitness levels and goals, offering personalized insights and recommendations. In the human recommendation condition, participants were informed that they had found a newly launched fitness app featuring recommendations from a human trainer who assesses their fitness goals and provides personalized guidance (see Supplemental Material B).

To assess participants' willingness to follow the guidance based on the recommendation source, they indicated how likely they were to accept workout advice from the app using a 7-point scale (1 = not likely at all; 7 = very likely). To evaluate perceptions of the recommendation's objectivity, participants also rated how objective or subjective the advice seemed (e.g., “The workout advice you receive from the AI/human trainer is ...”) on a 7-point scale (1 = completely objective; 7 = completely subjective). Finally, demographic information, including age and gender, was collected.

4.2. Results and discussion

First, the analysis of the mindset manipulation check confirmed a significant difference between deliberative ($M = 4.51$, $SD = 1.52$) and implemental mindsets ($M = 5.31$, $SD = 1.62$; $t(246) = -4.03$, $p < .001$, $d = -0.51$, 95% CI = $[-1.20, -0.41]$).

A 2×2 ANOVA was conducted to examine whether the intention to accept recommendations from a human trainer versus an algorithm differed between participants in deliberative and implemental mindsets. Results revealed a significant interaction effect between mindset and recommendation type ($F(1, 244) = 9.05$, $p = .003$; $\eta_p^2 = 0.04$). The main effect of mindset was not significant ($F(1, 244) = 0.09$, $p = .761$), and the main effect of recommendation type was also not significant ($F(1, 244) = 0.18$, $p = .672$). Post-hoc analyses revealed that participants in a deliberative mindset were marginally more likely to accept recommendations from a human trainer ($M = 5.28$, $SD = 1.42$) compared to those from an algorithm ($M = 4.77$, $SD = 1.80$; $F(1, 244) = 3.63$, $p = .058$, 95% CI = $[-1.03, 0.02]$; $\eta_p^2 = 0.02$). However, participants in an implemental mindset were more likely to accept recommendations from an algorithm ($M = 5.42$, $SD = 1.34$) compared to a human trainer ($M = 4.75$, $SD = 1.52$; $F(1, 244) = 5.45$, $p = .020$, 95% CI = $[0.11, 1.24]$; $\eta_p^2 = 0.02$; see Fig. 1).

This study also measured perceived objectivity and found a non-significant interaction effect between mindset and recommendation type ($F(1, 244) = 0.93$, $p = .337$). The main effect of mindset was marginally significant ($F(1, 244) = 3.30$, $p = .070$; $\eta_p^2 = 0.01$), while the main effect of recommendation type was strongly significant ($F(1, 244) = 13.89$, $p < .001$; $\eta_p^2 = 0.05$). Post-hoc analyses revealed that participants in both deliberative and implemental mindsets perceived recommendations from an algorithm ($M_{\text{deliberative}} = 3.65$, $SD = 1.62$ vs. $M_{\text{implemental}} = 3.82$, $SD = 1.75$; $F(1, 244) = 0.35$, $p = .553$) as more objective compared to recommendations from a human trainer

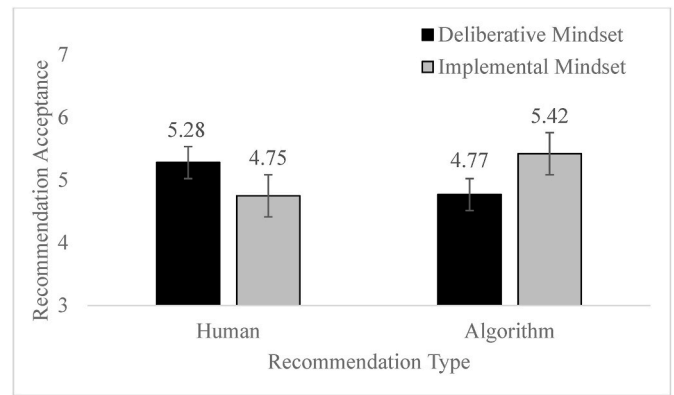


Fig. 1. Mean acceptance of recommendations in Study 1A. Note. Error bars represent ± 1 SE.

($M_{\text{deliberative}} = 4.20$, $SD = 1.54$ vs. $M_{\text{implemental}} = 4.76$, $SD = 1.39$; $F(1, 244) = 3.99$, $p = .047$, 95% CI = $[0.01, 1.11]$; $\eta_p^2 = 0.02$).

This study examined how deliberative and implemental mindsets shape individuals' acceptance of recommendations from human experts versus algorithms. A significant interaction emerged between mindset and recommendation source in predicting acceptance intentions. Among those in a deliberative mindset, responses varied by recommendation type at a marginal level of significance, suggesting a potential attentiveness to the source of the recommendation. Additionally, participants generally perceived algorithmic recommendations as more objective than those from human experts regardless of mindset, supported by a significant main effect of recommendation type, while effects involving mindset and the interaction were less pronounced.

To further deepen our understanding of these effects, it is essential to replicate the results and explore the specific situations in which individuals exhibit a preference for recommendations from human experts over algorithms. In this regard, Study 1B was conducted in a different context, employing a new dependent variable to directly assess participants' preferences for each recommendation type.

5. Study 1B

Study 1B aimed to replicate the findings of Study 1A while testing the main hypothesis in a new context—planning a trip to South America. This scenario was selected to provide a distinct yet relatable decision-making setting, allowing for an examination of the robustness and generalizability of the effects observed in the previous workout scenario. It was hypothesized that individuals in a deliberative mindset would respond more favorably to human-generated recommendations, whereas those in an implemental mindset would respond more favorably to algorithm-generated recommendations.

5.1. Method

Study 1B targeted 400 participants and received responses from 386 U.S.-based participants through Prolific ($M_{\text{age}} = 40.61$; 49.0% female,

49.7% male, and 1.3% other).⁴ A post hoc power analysis conducted with G*Power revealed that the study achieved 57% power to detect the observed effect size. Randomization checks revealed no significant differences in age ($p = .920$) or gender ($p = .773$) across conditions. Participants were randomly assigned to a 2-cell (mindset: deliberative versus implemental) between-subjects design.

The mindset manipulation was identical to that used in Study 1A. After the manipulation, participants were asked to imagine traveling to South America to learn a new culture and take a Spanish course at a language school. They were informed that they would need to choose a city and that the language school's website offered two tools to help them: an algorithm and a human expert. To assess participants' explicit preferences and behavioral intentions regarding reliance on algorithmic versus human advice, they rated their preferred tool on a scale from 1 (definitely the algorithm tool) to 7 (definitely the human expert tool). Additionally, participants reported their intentions to follow algorithm-based recommendations (e.g., "How much do you intend to follow algorithm-based recommendations in making your choice?") on a scale from 1 (not at all) to 7 (very much) (see Supplemental Material C). Finally, demographic information, including age and gender, was collected.

5.2. Results and discussion

An independent samples t -test revealed that participants in a deliberative mindset ($M = 5.31$, $SD = 1.72$) expressed a stronger preference for recommendations from human experts than those in an implemental mindset ($M = 4.91$, $SD = 1.86$; $t(384) = 2.18$, $p = .030$, $d = 0.22$, 95% CI

$= [0.39, 0.76]$; see Fig. 2). Additionally, participants in an implemental mindset ($M = 4.40$, $SD = 1.55$) were more inclined to follow algorithm-based recommendations than those in a deliberative mindset ($M = 4.06$, $SD = 1.52$; $t(384) = -2.14$, $p = .033$, $d = -0.22$, 95% CI $= [-0.64, -0.03]$).

The results of Study 1B supported H1A and H1B, showing that individuals in a deliberative mindset were more likely to prefer recommendations from human experts, while those in an implemental mindset exhibited a stronger preference for algorithmic advice. These findings highlight the influence of mindset on recommendation preferences; however, the context of decision-making is far from straightforward. For instance, when using a newly launched fitness app to achieve health goals, the uncertainty surrounding the app's reliability and effectiveness may amplify concerns. Such perceived uncertainty may influence how individuals evaluate and respond to algorithmic versus human recommendations. To investigate this further, Study 2 and a related pretest examined how uncertainty moderates preferences for algorithmic versus human recommendations, particularly across deliberative and implemental mindsets.

6. Study 2

Study 2 aimed to examine whether perceived uncertainty moderates the influence of deliberative versus implemental mindsets on preferences for algorithmic versus human recommendations. Specifically, the study proposed that under high uncertainty, participants in a deliberative mindset would prefer recommendations from human experts over algorithmic recommendations, while those in an implemental mindset would favor algorithmic recommendations over human expert recommendations. However, under low uncertainty, this effect was expected to reverse.

6.1. Method

This study aimed to recruit 400 participants through Prolific. Participants who did not correctly answer the attention check question, as in Study 1A, were excluded from the analysis, resulting in a final sample of 386 U.S.-based participants ($M_{age} = 44.06$; 49.0% female, 49.7% male, and 1.3% other). A post hoc power analysis conducted with G*Power revealed that the study achieved 69% power to detect the observed effect size. Randomization checks revealed no significant differences in age ($p = .643$) or gender ($p = .800$) across conditions. A 2 (mindset: deliberative versus implemental) $\times$ 2 (uncertainty: high versus low) between-subjects design was used, with participants randomly assigned to one of the four conditions.⁵

First, the manipulation of deliberative and implemental mindsets, along with the manipulation check question, was identical to Study 1A. After completing the mindset manipulation, participants encountered a scenario involving investment decisions (see Supplemental Material D). This scenario was chosen because investment decisions inherently involve varying levels of uncertainty, which allows for experimental manipulation and examination of how different uncertainty contexts interact with mindset to affect preferences for algorithmic versus human recommendations.

In the high uncertainty condition, participants imagined considering

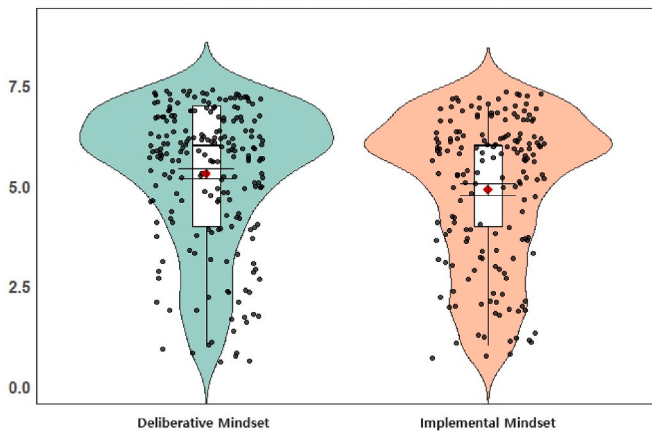


Fig. 2. Mean preference for human recommendations in Study 1B.

⁴ Due to an unexpected error during the process of collecting data from 400 participants, the study began by recruiting 301 participants through Prolific, who completed the study in exchange for nominal monetary compensation. Participants who did not correctly answer the attention check question, as in Study 1A, were excluded from the analysis, resulting in a final sample of 288 participants. The analysis revealed that participants in a deliberative mindset ($M = 5.25$, $SD = 1.74$) were more likely to prefer recommendations from human experts than those in an implemental mindset ($M = 4.80$, $SD = 1.89$; $t(286) = 2.11$, $p = .036$, $d = 0.25$, 95% CI $= [0.03, 0.88]$). To reach the intended sample size, data collection was resumed, and an additional 99 participants were subsequently recruited (Importantly, this extension was prompted by a procedural issue rather than the significance of initial findings). The effect on the main DV (preference) remained stable. Notably, the effect on intention to follow, initially nonsignificant, reached significance in the combined sample, suggesting that the change was due to increased statistical power rather than selective analysis.

⁵ A pretest was conducted to identify which domain would be perceived as involving the most uncertainty. This pretest examined three domains previously used in Study 1A (Using a newly launched fitness app on the market to achieve health goals), Study 1B (traveling in South America), and Study 2 (investing in a high-risk stock market). Participants rated how uncertain they felt about each situation on a scale from 1 (not uncertain at all) to 7 (very uncertain). A total of 300 participants were recruited via Prolific, and a 3-cell between-subjects design was employed. The results revealed no significant difference between the three conditions ($F(1, 297) = 1.43$, $p = .242$).

investing their savings in the stock market during a period of heightened volatility, marked by rapid fluctuations driven by global economic instability and unforeseen events. Conversely, in the low uncertainty condition, participants imagined considering investing during a period of relative market stability. To create clear distinctions between mindsets and reduce potential issues with skewed distributions, this study used a binary DV. Participants indicated their preference between an algorithmic tool and a human expert tool for their investment (0 = definitely the algorithmic tool; 1 = definitely the human expert tool).

To enhance the rigor and clarity of the study, participants' pre-existing trust in algorithmic recommendations was explicitly measured and statistically controlled for during the analysis (e.g., "In general, how much do you trust algorithm-based recommendations?").⁶ Including this control variable aimed to isolate the effects of mindset and uncertainty on recommendation preferences, ensuring that observed differences were not confounded by individual variations in trust. Finally, demographic information, including age and gender, was collected.

6.2. Results and discussion

First, the analysis of the mindset manipulation check confirmed a significant difference between deliberative ($M = 4.12$, $SD = 1.73$) and implemental mindsets ($M = 5.63$, $SD = 1.43$; $t(384) = -9.29$, $p < .001$, $d = -0.95$, 95% CI = $[-1.83, -1.19]$).

A binary logistic regression analysis was conducted to examine the effects of mindset and uncertainty on the preference for recommendations, controlling for pre-existing trust in algorithms. The covariate, pre-existing trust in algorithms, was significant ($B = -0.89$, $SE = 0.11$, $\text{Exp}(B) = 0.41$, $p < .001$, 95% CI = $[0.33, 0.51]$), indicating that higher levels of pre-existing trust were associated with a greater preference for recommendations from algorithms. After accounting for this covariate, the main effects of both mindset ($B = 0.74$, $SE = 0.34$, $\text{Exp}(B) = 2.11$, $p = .029$, 95% CI = $[1.08, 4.11]$) and uncertainty ($B = 0.75$, $SE = 0.36$, $\text{Exp}(B) = 2.11$, $p = .040$, 95% CI = $[1.04, 4.29]$) were significant. Additionally, a significant interaction effect between mindset and uncertainty emerged ($B = -1.11$, $SE = 0.49$, $\text{Exp}(B) = 0.33$, $p = .023$, 95% CI = $[0.13, 0.86]$).

Post-hoc analyses revealed that, after controlling for pre-existing trust, participants in a deliberative mindset were more likely to prefer recommendations from human experts (65.4%) than those in an implemental mindset in the high uncertainty condition (58.2%; $\chi^2(1) = 4.75$, $\text{Exp}(B) = 0.48$, $p = .029$, 95% CI = $[0.24, 0.93]$). However, in the low uncertainty condition, participants in an implemental mindset showed a greater inclination to prefer recommendations from human experts (67.0%) compared to those in a deliberative mindset, although this difference was not statistically significant (58.3%; $\chi^2(1) = 1.13$, $\text{Exp}(B) = 1.45$, $p = .289$, 95% CI = $[0.73, 2.88]$; see Fig. 3).

This study examined the significant interaction between mindset and uncertainty, revealing that under high uncertainty, individuals in an implemental mindset (vs. a deliberative mindset) showed a greater

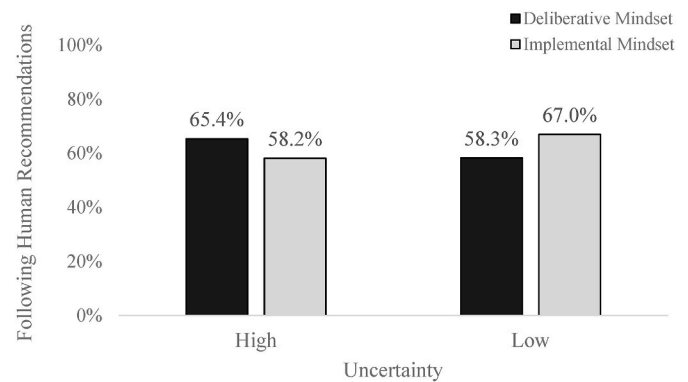


Fig. 3. Mean influence of mindset and uncertainty on the preference for recommendations from human experts in Study 2.

preference for algorithm-generated over human-generated recommendations. In contrast, under low uncertainty, the expected trend emerged, with implemental-minded individuals tending to prefer human-generated over algorithm-generated recommendations, although this difference was not statistically significant.

7. Study 3

Study 3 aimed to examine the effect of deliberative versus implemental mindsets on algorithm acceptance in an incentive-compatible setting, enhancing the validity of the findings. Specifically, the study investigated how the mindset an individual adopts influences their willingness to follow algorithmic recommendations, particularly when real incentives are involved. It was hypothesized that individuals in a deliberative mindset would respond more favorably to human-generated recommendations, whereas those in an implemental mindset would respond more favorably to algorithm-generated recommendations.

7.1. Method

Study 3 targeted 260 participants and recruited 257 U.S.-based participants through CloudResearch ($M_{\text{age}} = 38.79$; 50.2% female, 49.4% male, and 0.4% other). A post hoc power analysis conducted with G*Power revealed that the study achieved 36% power to detect the observed effect size.⁷ Randomization checks revealed no significant differences in age ($p = .188$) or gender ($p = .611$) across conditions. A 2 (mindset: deliberative versus implemental) $\times$ 2 (recommendation type: algorithm versus human) between-subjects design was used, with participants randomly assigned to one of the four conditions. Preregistration details can be found at <https://aspredicted.org/gmtd-npzn.pdf>.

The mindset manipulation and manipulation check were identical to those used in Study 1A. Following the initial task, participants answered a stock value estimation question based on the Judge Advisor System (JAS) paradigm, adapted from Logg et al. (2019), to assess the extent to which they assimilate advice from different sources (Sniezek & Buckley, 1995). In the JAS paradigm of Logg et al. (2019), participants first made a judgment in a certain context, then received external input "advice," and finally had the opportunity to adjust their initial assessment. This paradigm was chosen because it accurately captures how individuals integrate external advice into their judgments, which is essential for

⁶ This decision was guided by prior research emphasizing the critical role of pre-existing trust in unfamiliar decision-making contexts, where trust serves as a heuristic to reduce cognitive burden and navigate high ambiguity (Hobson et al., 2023). Financial decisions are particularly relevant in this regard, as they often involve significant stakes, irreversible outcomes, and complex trade-offs, all of which increase uncertainty and emotional intensity. In such situations, individuals may rely more heavily on simple heuristics such as trust when evaluating decision aids, which can lead to uncritical acceptance or avoidance of algorithmic advice. This dynamic is especially relevant in the current study, as reliance on trust-based heuristics could potentially obscure the effect of mindset, which was the primary psychological mechanism of interest. In addition, the covariate for pre-existing trust was not included in Studies 1A, 1B, and 3, as these studies were designed to establish the foundational relationship between mindset and receptivity to algorithmic versus human recommendations without introducing controls that could obscure emerging patterns.

⁷ A post hoc power analysis revealed 36% power for this study; however, the pattern of effects was observed consistently across multiple studies with varying power levels. This convergence strengthens confidence in the robustness of the findings, although the limited power in some cases warrants caution in interpretation. Further discussion of these limitations and their implications is provided in the General Discussion section.

understanding how mindsets influence acceptance of algorithmic versus human recommendations.

Next, participants estimated the future value of the Hang Seng Index (HSI), which represents companies listed on the Hong Kong Stock Exchange (see Supplemental Material E). They were given key market information, including the HSI's current value, its value from one year ago, its highest and lowest points over the past year, and a graph depicting its one-year performance (see Fig. 4). Based on this data, participants estimated the HSI's value two months into the future.

Next, in the algorithmic recommendation condition, participants were informed that a leading investment firm had recently developed a computer algorithm capable of estimating future stock values with an accuracy rate of 87%, compared to the estimates of recent MBA graduates. In the human recommendation condition, participants were told that a leading investment advisor (human expert) from a financial firm had made the same estimate, with an accuracy rate of 87%, also compared to the estimates of recent MBA graduates. Participants were then informed that the [algorithm/human expert] predicted the HSI would be worth 17,500 points two months from now. With this information, participants had the option to either revise their original estimate or maintain it. To ensure an incentive-compatible setting, participants were informed that a random draw would select one participant from the survey whose estimate ranked among the top 5% in accuracy. This participant received a \$5 bonus payment. Finally, demographic information, including age and gender, was collected.

7.2. Results and discussion

First, the analysis of the mindset manipulation check confirmed a significant difference between deliberative ($M = 3.84$, $SD = 1.63$) and implemental mindsets ($M = 5.23$, $SD = 1.60$; $t(255) = -6.85$, $p < .001$, $d = -0.86$, 95% CI = $[-1.78, -0.99]$).

This study utilized the Judge Advisor System (JAS) paradigm (Logg et al., 2019) to examine how participants incorporated advice. The weight on advice (WOA) was calculated as the difference between the initial and revised judgment, divided by the difference between the initial judgment and the advice. A WOA of 0 indicated that a participant ignored the advice entirely, while a WOA of 1 signified that they fully adopted the advice, abandoning their initial judgment. A WOA between 0 and 1 indicated partial incorporation of the advice, whereas a WOA greater than 1 suggested an overreaction to the advice.

Based on the WOA calculation, a 2×2 ANOVA was conducted to examine whether the intention to adopt recommendations from a human expert versus an algorithm differed between participants in deliberative and implemental mindsets. Results revealed a marginally significant interaction effect between mindset and recommendation type ($F(1, 253) = 3.15$, $p = .077$; $\eta_p^2 = 0.01$). The main effect of mindset was not significant ($F(1, 253) = 0.14$, $p = .706$), and the main effect of recommendation type was also not significant ($F(1, 253) = 1.28$, $p =$

.259). Post-hoc analyses revealed that participants in a deliberative mindset were more likely to adopt recommendations from a human expert ($M = 0.54$, $SD = 0.55$) than recommendations from an algorithm ($M = 0.38$, $SD = 0.41$; $F(1, 253) = 4.48$, $p = .035$, 95% CI = $[0.01, 0.31]$; $\eta_p^2 = 0.02$). However, in an implemental mindset, participants exhibited a slight tendency to rely more on recommendations from an algorithm ($M = 0.49$, $SD = 0.38$) compared to a human expert ($M = 0.46$, $SD = 0.37$; $F(1, 253) = 0.20$, $p = .659$, 95% CI = $[-0.10, 0.16]$). Although this difference was not statistically significant, the pattern remained consistent.

This study examined how deliberative versus implemental mindsets influence the adoption of recommendations from human experts versus algorithms in an incentive-compatible setting. By introducing real stakes, the study aimed to capture more authentic decision-making behaviors that better reflect real-world contexts. The results suggested a directional, nuanced interaction: participants in a deliberative mindset were significantly more inclined to adopt recommendations from human experts, whereas those in an implemental mindset showed a tendency to prefer algorithmic recommendations, although this preference did not reach statistical significance. These findings provide subtle support for the idea that mindset shapes how individuals engage with recommendations and underscore the importance of considering psychological states when evaluating advice in high-stakes decision-making contexts.

8. General discussion

As algorithmic recommendations become increasingly embedded in everyday decision-making, understanding the psychological factors that shape how individuals respond to algorithmic input has become essential. While some studies highlight algorithm appreciation, where people prefer algorithmic input for its perceived objectivity and efficiency (Logg et al., 2019), others document algorithm aversion, where individuals favor human judgment even when algorithms outperform humans in accuracy (Castelo et al., 2019; Dietvorst et al., 2015). Much of this prior work has focused on the characteristics of the algorithm or the nature of the task, with less attention paid to the psychological state of the decision-maker. The present research addresses this gap by demonstrating that receptivity to algorithmic versus human recommendations depends, in part, on a person's cognitive orientation, specifically whether they are in a deliberative or implemental mindset, reflecting their readiness to act.

Across four studies, this work examined how deliberative and implemental mindsets shape individuals' preferences for algorithmic versus human recommendations. Study 1A supported H1A and H1B, revealing a significant interaction between mindset and recommendation source in predicting acceptance intentions. Individuals in an implemental mindset were more receptive to algorithmic recommendations, whereas those in a deliberative mindset demonstrated marginal sensitivity to the source. This pattern was replicated in Study 1B within a different decision-making context. These findings suggest that even similar recommendations may be interpreted differently depending on the decision-maker's psychological readiness to act, highlighting the nuanced role of mindset in shaping evaluative processes.

Study 2 extended these insights by examining uncertainty as a moderating variable, and the results supported H2A and H2B. Under high uncertainty, individuals in a deliberative mindset exhibited a stronger preference for human-generated recommendations, whereas those in an implemental mindset favored algorithmic advice. Under low uncertainty, this pattern reversed: individuals in an implemental mindset tended to prefer human over algorithmic recommendations, although this difference was directional and did not reach statistical significance. These results imply that the perceived usefulness of recommendation sources is influenced by the interplay between mindset and contextual uncertainty.

Study 3 introduced consequential stakes to better approximate real-

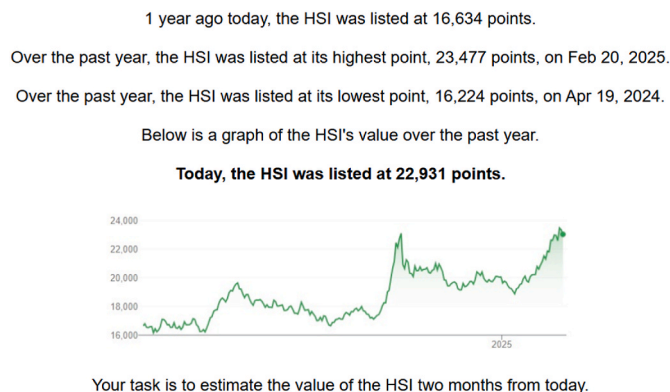


Fig. 4. Key information provided about the Hang Seng Index (HSI) in Study 3.

world decision environments. In an incentive-compatible setting, the results revealed a nuanced interaction: participants in a deliberative mindset were significantly more inclined to adopt recommendations from human experts, whereas those in an implemental mindset tended to adopt algorithmic recommendations, although this was not statistically significant. Overall, these findings underscore the subtle role of individuals' psychological states, particularly their mindset, in shaping how people interact with algorithmic versus human recommendations. By bridging insights from research on mindsets and algorithmic decision-making, this work sheds light on when and why people embrace or resist algorithmic input and suggests potential avenues for improving human–algorithm collaboration in practice.

8.1. Theoretical contributions

This research makes theoretical advances by integrating mindset theory (Gollwitzer, 1990) with the literature on human–algorithm interaction. By doing so, it offers a novel cognitive-motivational framework that helps explain variability in algorithm acceptance. Previous studies on algorithm aversion and appreciation primarily emphasized static individual differences (e.g., trust, technology readiness, personality traits) or external system characteristics (e.g., transparency, error rates) (Dietvorst et al., 2015; Logg et al., 2019; Mahmud et al., 2022). However, these approaches have struggled to explain the dynamic fluctuations in people's receptiveness to algorithmic advice observed across different decision contexts and phases. By emphasizing psychological readiness to act, as reflected in deliberative versus implemental mindsets, this research advances theory by showing that shifts in cognitive orientation over time can meaningfully alter how individuals process information and, in turn, shape their openness to algorithmic versus human input.

This theoretical synthesis refines cognitive-motivational frameworks by suggesting that motivation operates as a foundational mechanism shaping how individuals engage in epistemic evaluation throughout the decision-making process. It challenges static conceptions of information acceptance by revealing how dynamic motivational states, manifested through deliberative and implemental mindsets (Gollwitzer, 1990), govern the appraisal and integration of algorithmic versus human recommendations. These motivational dimensions influence cognitive engagement and selective openness, aligning information processing with evolving goals and readiness to act (Lee & See, 2004; Metzger & Flanagin, 2013). By centering motivation in the fluid process of epistemic evaluation, this work offers a transformative lens that deepens our understanding of variability in technology acceptance and broadens cognitive-motivational theories of adaptive decision behavior.

Furthermore, the interplay between mindset and perceived uncertainty enriches theoretical insight into the complex contextual moderators shaping algorithm acceptance. While prior research has acknowledged uncertainty as a significant factor influencing reliance on algorithms (Dietvorst & Bharti, 2020; Perkins et al., 2010; Zhang et al., 2021), this study advances understanding by elucidating the underlying cognitive mechanisms through which uncertainty modulates the effects of motivational states on decision-making processes. Specifically, it reveals how heightened uncertainty intensifies the differential processing styles associated with deliberative and implemental mindsets, thereby amplifying their impact on openness to algorithmic versus human recommendations. Conversely, reduced uncertainty diminishes these mindset-driven distinctions, highlighting the fluid and contingent nature of cognitive engagement in algorithmic decision contexts. This nuanced perspective extends cognitive-motivational theory by incorporating situational factors, underscoring that motivational influences on technology acceptance are contingent on the evolving informational environment.

8.2. Practical implications

The findings of this study have important implications for marketers, policymakers, and organizations that develop algorithmic decision-support systems. Understanding the psychological mechanisms underlying individual preferences for algorithmic versus human recommendations can inform strategies to enhance algorithm acceptance. Marketers may optimize the presentation of algorithmic recommendations by tailoring messaging to align with consumers' mindset. For instance, emphasizing the action-oriented benefits of algorithmic recommendations can enhance acceptance among consumers in an implemental mindset, while highlighting the thoroughness and personalized insights of human recommendations may resonate more with those in a deliberative mindset.

Furthermore, firms can create recommendation aids designed to guide consumers in understanding the optimal contexts for using algorithmic tools. By providing consumers with clear guidance on when it is more appropriate to rely on algorithmic solutions and when to seek human expertise, businesses can promote a more balanced and informed decision-making process. This approach is especially valuable in business sectors, such as healthcare and legal services, where algorithmic decision aids are becoming an integral part of professional advisory systems. Through this strategy, firms can ensure that consumers feel empowered to make decisions that are both data-driven and human-centered, ultimately enhancing overall reliance on the recommendations provided.

8.3. Limitations and directions for future research

Several limitations deserve attention. First, although this study experimentally manipulated psychological readiness using deliberative and implemental mindsets, it did not examine how these mindsets are naturally elicited in real-world settings. In everyday life, a deliberative mindset may emerge during initial planning phases when individuals weigh multiple options, whereas an implemental mindset is more likely to arise after a decision has been made and action is underway. Future research should focus on identifying the situational cues and decision contexts that trigger these mindsets organically, thereby improving the ecological validity and applicability of laboratory findings.

Second, power analyses were conducted post hoc across all studies, and some studies exhibited relatively low statistical power. While the observed patterns were consistent, the effect sizes varied and were not uniformly strong. Additionally, some findings reached only marginal significance, indicating trends that warrant cautious interpretation rather than definitive conclusions. To enhance the reliability and generalizability of these results, future research should aim to replicate the findings with larger and more diverse samples.

While the study identifies a clear mechanism linking motivational states to receptivity to algorithmic versus human recommendations, it offers a nuanced perspective, emphasizing that the findings are neither uniform nor guaranteed across all contexts. Adoption and trust are dynamic and fluid, shaped by the interplay between internal psychological states and external decision environments. Trust itself can fluctuate over time, further underscoring the conditional nature of algorithmic acceptance. Accordingly, these findings should be interpreted with careful attention to context and boundary conditions, recognizing that receptivity to algorithmic input may vary across different influencing factors.

Moreover, this work measured only age and gender as demographic variables, but it is crucial to examine additional potential moderating factors, such as task complexity, individual differences in technology familiarity, and cultural background, which may influence the strength or direction of these effects. By differentiating robust from tentative findings and exploring boundary conditions, future studies can refine theoretical models and provide more precise guidance for the design and implementation of algorithmic decision-support systems.

Relatedly, pre-existing trust was measured and controlled only in Study 2, following prior research highlighting its critical role in unfamiliar decision-making contexts where such trust acts as a heuristic to reduce cognitive burden and navigate high ambiguity (Hobson et al., 2023). Future studies should investigate whether this effect is universal across diverse populations and decision settings. Additionally, research should explore effective interventions to strengthen trust in algorithms and thereby enhance individuals' openness to algorithmic recommendations.

The present work primarily focused on the source of recommendations, human versus algorithm, without manipulating key algorithmic attributes such as explainability. These factors could significantly influence user trust and acceptance of algorithmic systems. For instance, increased explainability could help individuals in a deliberative mindset overcome concerns about algorithms' perceived rigidity or impersonal nature by offering clearer insights into how recommendations are generated. Future research should examine how such algorithmic features interact with psychological states like mindset and contextual factors such as uncertainty to influence both trust and reliance on algorithms.

In conclusion, this research advances the understanding of psychological drivers underlying algorithm appreciation and aversion by identifying mindset as a critical factor. By illuminating how mindset shapes receptiveness to algorithmic advice, the study opens new pathways for improving the design and communication of algorithmic systems. Future research could investigate how tailoring algorithmic interfaces and interactions to users' motivational states enhances effectiveness and acceptance, thereby promoting a more seamless and synergistic integration of human and algorithmic decision-making in everyday life.

CRediT authorship contribution statement

Hyoseok Kim: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Informed consent

Informed consent was obtained from all participants included in the studies.

Artificial intelligence

The author used generative AI to ensure proofreading accuracy.

Ethical approval and statement

All procedures used in studies involving human participants are in accordance with the tenets of the Declaration of Helsinki or comparable ethical standards. This study was approved by the author's institution IRB (Approval no. #916) on June 26, 2023.

Declaration of generative AI

The author used generative AI (ChatGPT 4.0) to ensure proofreading accuracy. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Funding sources

This research was supported by the SCSU MRRC research grant.

Declaration of competing interest

The author declares that he has no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author has no acknowledgements to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2025.108824>.

Data availability

I have included the link to my data and code in the manuscript.

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